Disease Prediction from Medical Data

CodeAlpha Machine Learning Internship - Task 4

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Dataset: Breast Cancer Wisconsin Diagnostic Dataset

Domain: Machine Learning

Abstract

This project develops a machine learning based disease prediction system using structured medical data. The model predicts whether a breast tumor is malignant or benign. Multiple classification algorithms are trained and compared using standard evaluation metrics.

Objective

The objective is to predict the possibility of disease using medical features and classification models such as Logistic Regression, Support Vector Machine, Decision Tree, and Random Forest.

Dataset Description

Property	Value
Dataset	Breast Cancer Wisconsin Diagnostic
Records	569
Input Features	30
Target	Diagnosis
Classes	Malignant and Benign

Methodology

- Load and inspect the medical dataset.
- Remove duplicates and handle missing values.
- Encode the target classes.
- Split the data into training and testing sets.
- Standardize input features for Logistic Regression and SVM.
- Train Logistic Regression, SVM, Decision Tree, and Random Forest models.
- Compare models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- Save the best model and preprocessing objects.

Model Results

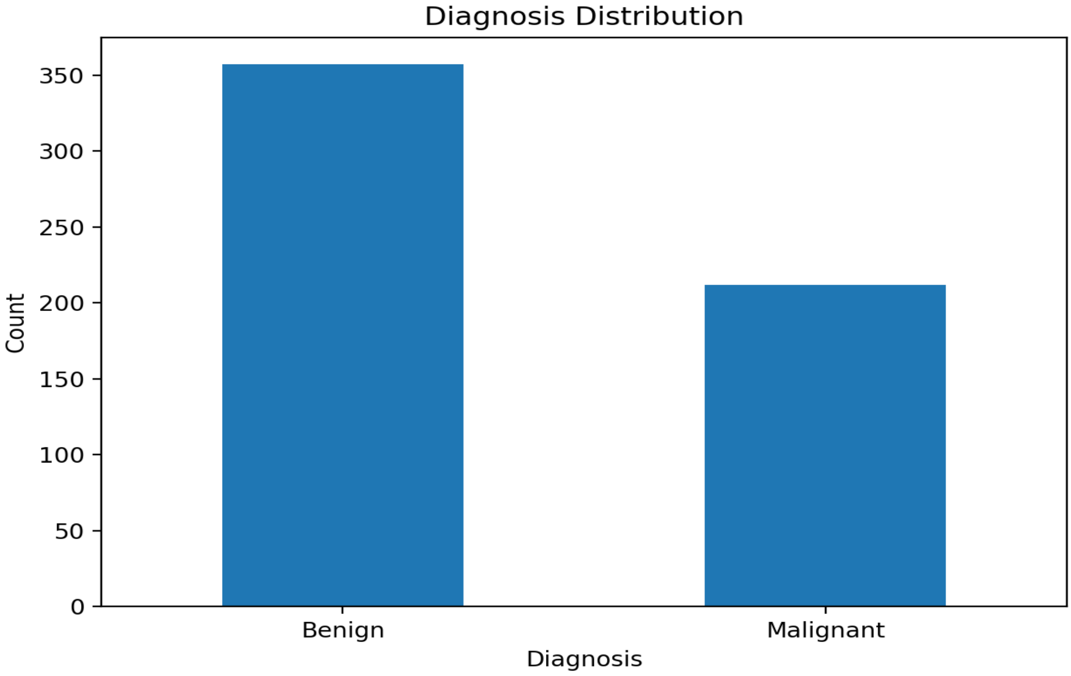
Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.9825	0.9861	0.9861	0.9861	0.9954
SVM	0.9825	0.9861	0.9861	0.9861	0.9950

Decision Tree	0.9123	0.9559	0.9028	0.9286	0.9157
Random Forest	0.9561	0.9589	0.9722	0.9655	0.9931

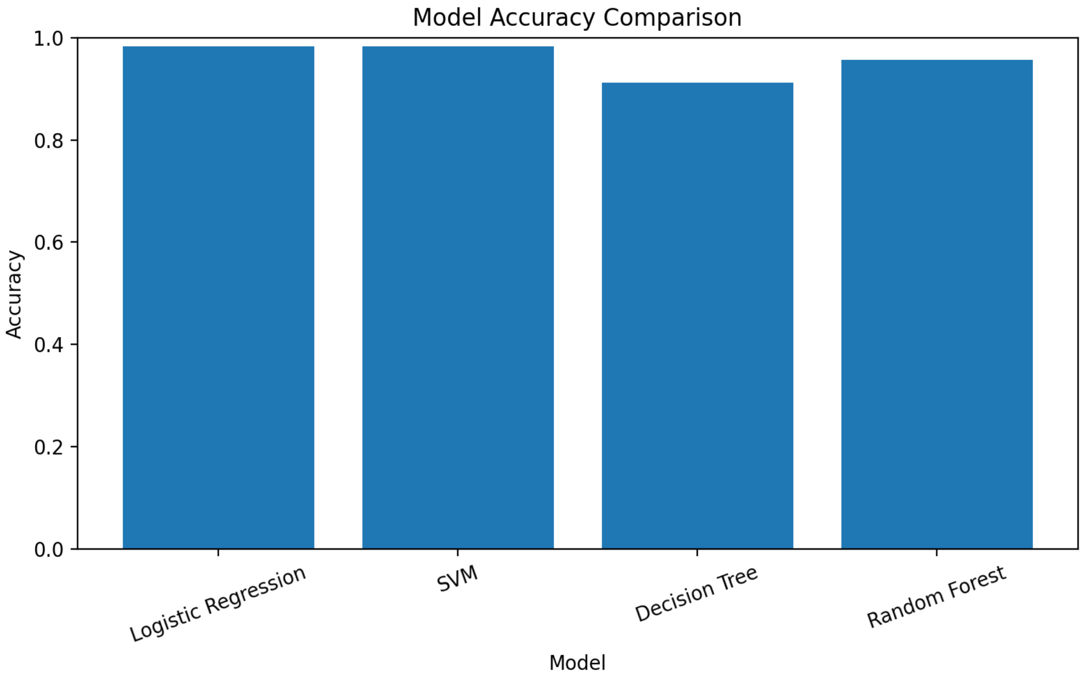
Best Model: Logistic Regression

Project Visualizations

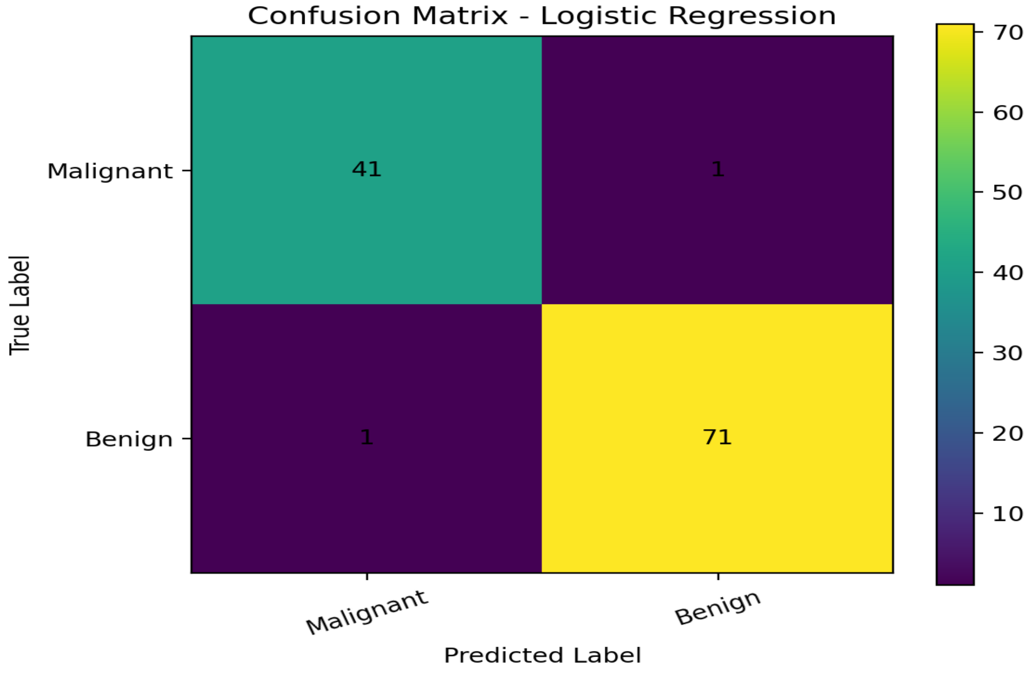
Diagnosis Distribution



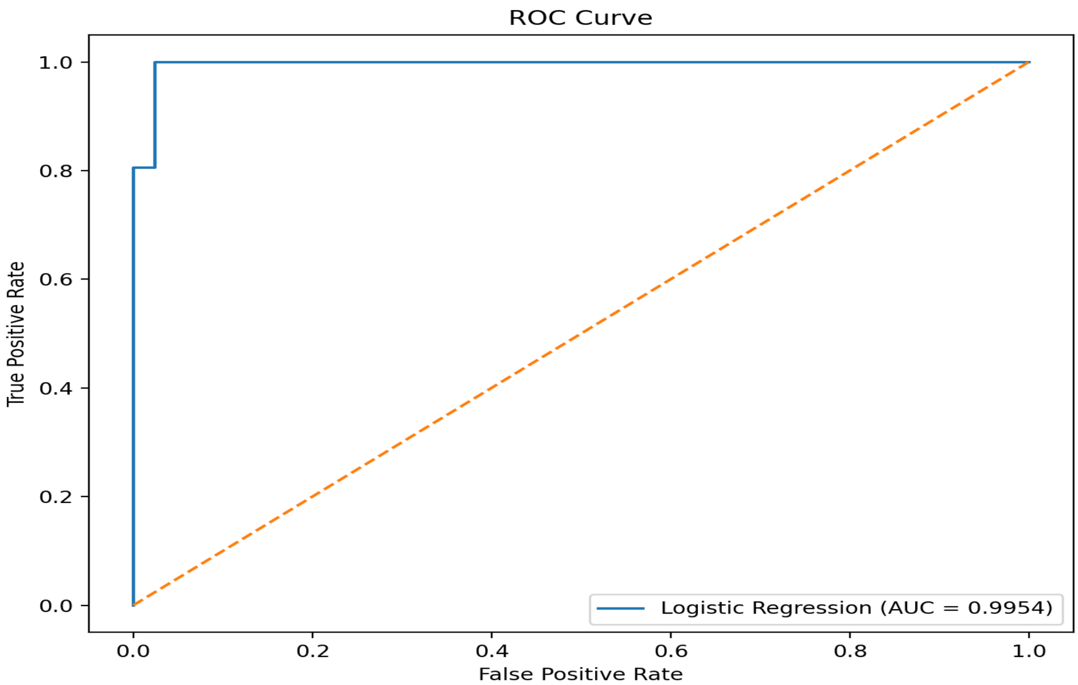
Model Accuracy Comparison



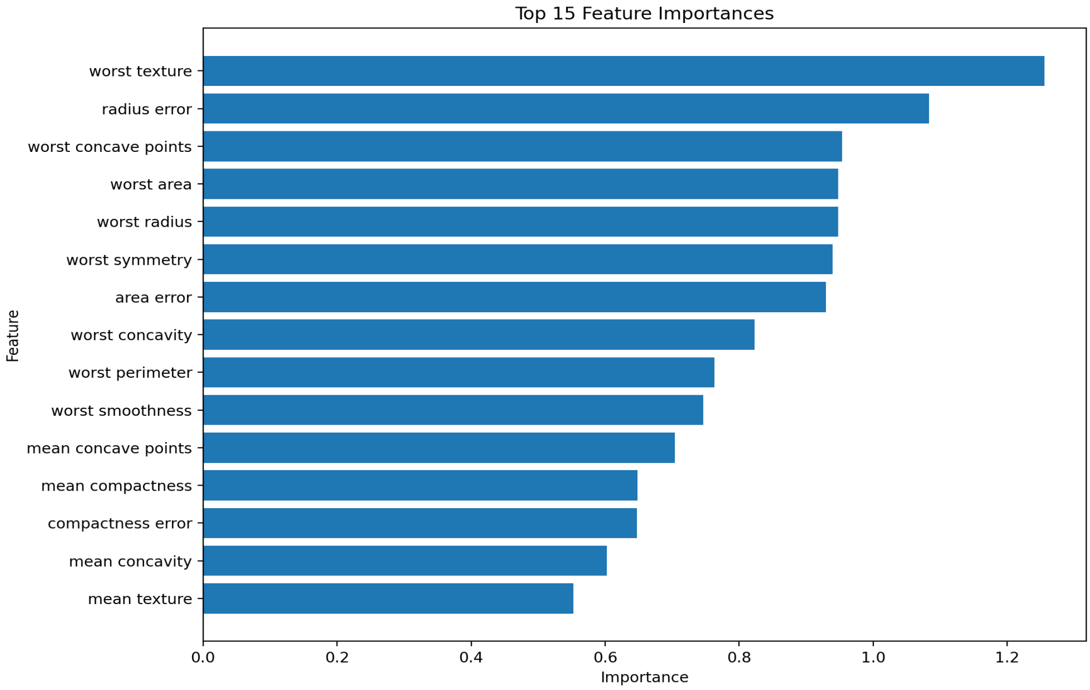
Confusion Matrix



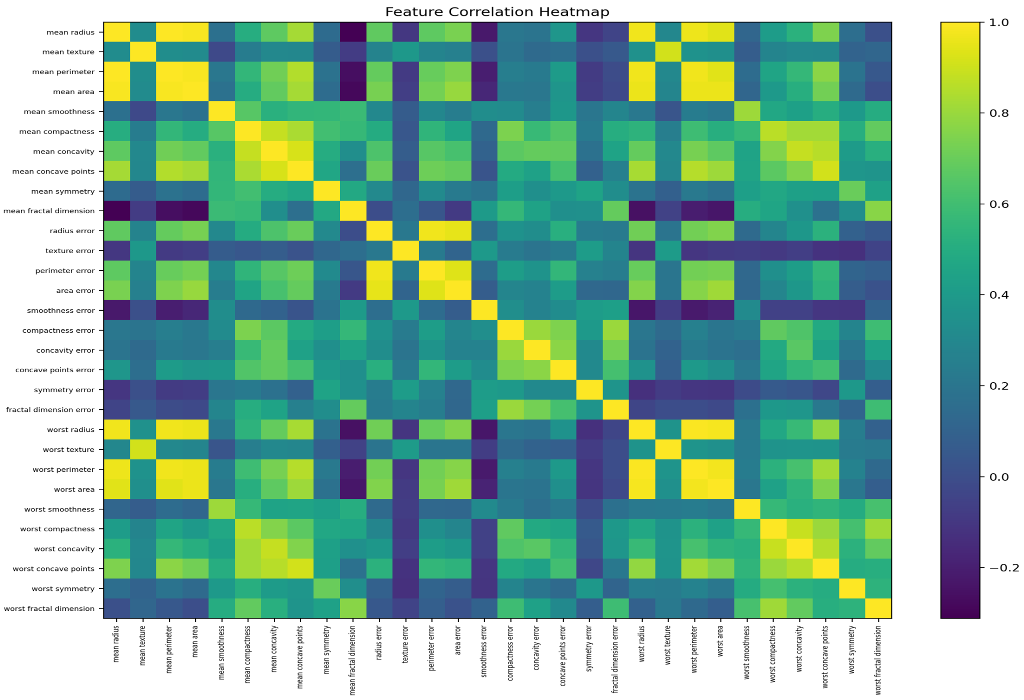
ROC Curve



Feature Importance



Correlation Heatmap



Conclusion

The project successfully demonstrates disease prediction from structured medical data. The trained models achieved strong classification performance, and the best model was saved for future prediction. This project is intended only for educational purposes and should not be used as a substitute for professional medical diagnosis.

Future Scope

- Use a larger real-world clinical dataset.

- Add XGBoost and hyperparameter tuning.

- Deploy the model using Streamlit or Flask.
- Add explainable AI methods such as SHAP.
- Build a secure patient-data prediction interface.